

V. Narayan Kushwaha

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EDUCATION

University at Buffalo, State University of New York

B.S. Aerospace Engineering (Honors) and B.S. Mechanical Engineering (Honors)

GPA: 3.3/4.0 | Dean's List: Fall 2023, Spring 2024

WORK EXPERIENCE

Lead Staff | SEAS DREAM Lab Makerspace

March 2024 – Present | Buffalo, NY

- Direct procurement, EH&S compliance, workshop scheduling and operations supporting students across FDM 3D printing, CNC machining, laser cutting, electronics and hybrid DREAM learning modules.
- Maintain Prusa 3D printer fleet (MK4S, MK3S, MK3, MINI); improved 3D print reliability by 40% through firmware debugging and thermal calibration and testing.
- Conducted 30+ hands-on workshops covering Arduino, CAD workflows, 3D printing optimization, and outreach/teaching
- Designed and deployed NFC-based print-logging system, reducing queue delays and filament waste by 25%
- Programmed and deployed a 6-Axis UFactory 850 Robotic arm for automatic 3D print retrieval and loading upon completion.

STEM Tutor | UB Tutoring and Academic Support Services

Jan 2024 – Present | Buffalo, NY

- Tutored 200+ students one-on-one in General Chemistry, Physics I–II, and Calculus I–III using MATLAB, Desmos, and PhET simulations for enhanced comprehension.

Engineering Intern | Sketch Consultants

May 2025 – Aug 2025 | Lucknow, India

- Developed python scripts to automate beam-load calculations and structural-analysis workflows, reducing redundant manual calculations by 40%
- Designed cabinets and equipment enclosures in FreeCAD using DfM principles; modeled steady-state heat conduction through building materials
- Conducted site inspections and coordinated with to validate with the architect's CAD models through on-site LiDAR Mapping and Photogrammetry.

RESEARCH EXPERIENCE

Undergraduate Researcher | UB DRONES Lab – Dr. Karthik Dantu

Aug 2026 – Present

VISTA : Vegetation Informed Sensing Through Active Mapping

EAGLE : Energy-Aware Guidance and Flight Estimation

- Processed and analyzed Pixhawk/ArduPilot telemetry and battery data.
- Built and compared physics-based, machine-learning, and hybrid energy models for energy-aware path planning.
- Collaborated with a multidisciplinary team to integrate aerial robotics and remote sensing technologies.
- Piloting a Hexacopter Drone with a FLIR Hyperspectral camera stack for data collection & crop surveillance.

Undergraduate Researcher, 3D Metal Printing Process Monitoring

UB Composite Materials Research Lab | Dr. Deborah Chung

Oct 2023 – Nov 2024

- Conducted transient electrical monitoring of metal additive manufacturing using an LCR scope to calculate fiber's electrical properties
- Measured inductance, resistance, impedance, and phase angle across varying frequencies on 8μ Boron Fiber samples
- Analyzed frequency domain electrical signals to identify fiber microstructure anomalies; co-authored conference paper accepted to MS&T 2024 Conference at Pittsburgh, PA

ENGINEERING PROJECT EXPERIENCE

Modular Surveillance VTOL UAV | In-Progress Flagship Senior Design Project

- Integrating VTOL flight, FPV, Thermal/NV imaging, GPS navigation, autonomous loiter, and LoRa telemetry and commands for route planning, Drone-like BLOS vertical takeoff/landing capability.
- Leading a 4-person design group, including software, structures, avionics, and aerodynamics subteams.
- Developed a computational design workflow using nTop for optimization, ANSYS Fluent for aerodynamic and stability analysis, and Fusion 360 for structural and assembly design

Pipe Flow & FEA Solver | MATLAB, Finite Element Analysis, Optimization

- Built MATLAB-based finite-element solvers and pipe-network analysis tools for design optimization and thermal performance prediction, HDPE & Steel pipe network optimization and steady-state heat-conduction analysis for heat exchangers and heat pumps.

Polargraph CNC Plotter | SOLIDWORKS, Arduino, Kinematics, Motion Control

- Designed a cable driven wall CNC plotter combining inverse kinematics and synchronized stepping in polar coordinates, Implemented Arduino motion control with Manual Stepping, SVG and G-code input.

CERTIFICATIONS

MATLAB Machine Learning Onramp | Simulink Control Design Onramp | Computer Vision Onramp | Siemens NX Xcelerator Academy | nTop Generative Design | MLH Best Hardware Hack | First-Year Engineering Design Award

SOFTWARES

MATLAB, Simulink, Simscape, SOLIDWORKS, Siemens NX, Fusion 360, FreeCAD, Python, C/C++, Arduino, ROS, OpenCV, Git, LaTeX, Microsoft Office